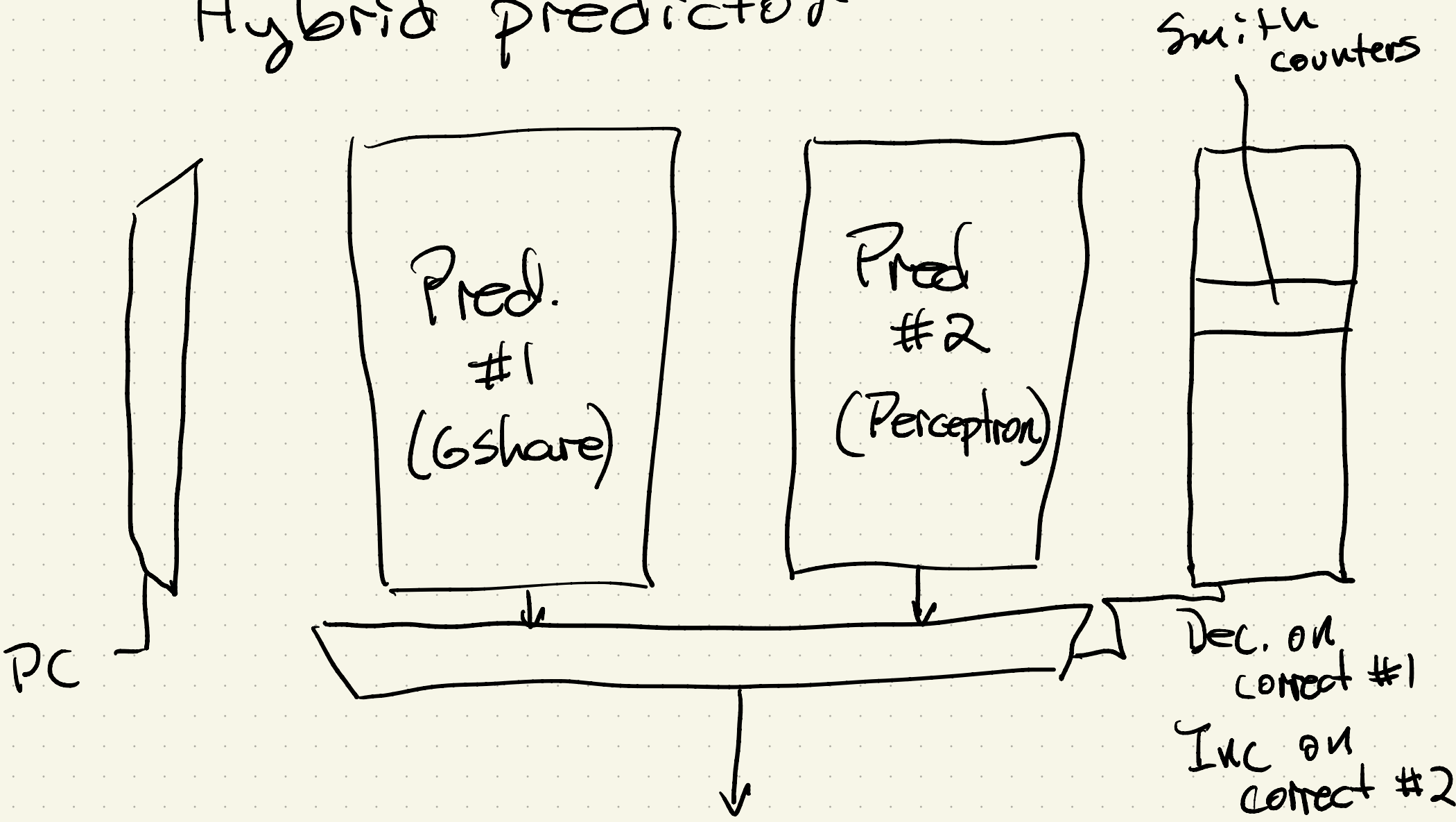
Hybrid predictor



Pipeline depth ↑ means need higher prediction accuracy

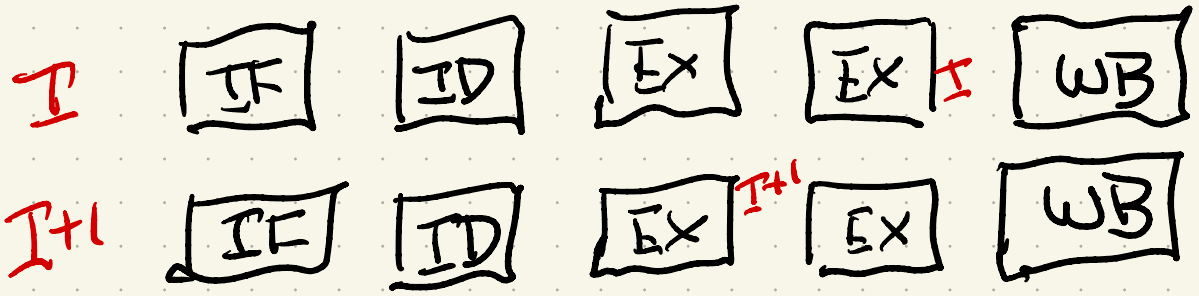
TAGE

IC * CPI * CT

low CT, CPI ≈ 1

CPI < 1 ? → (CPI)$^{-1}$ > 1

IPC > 1

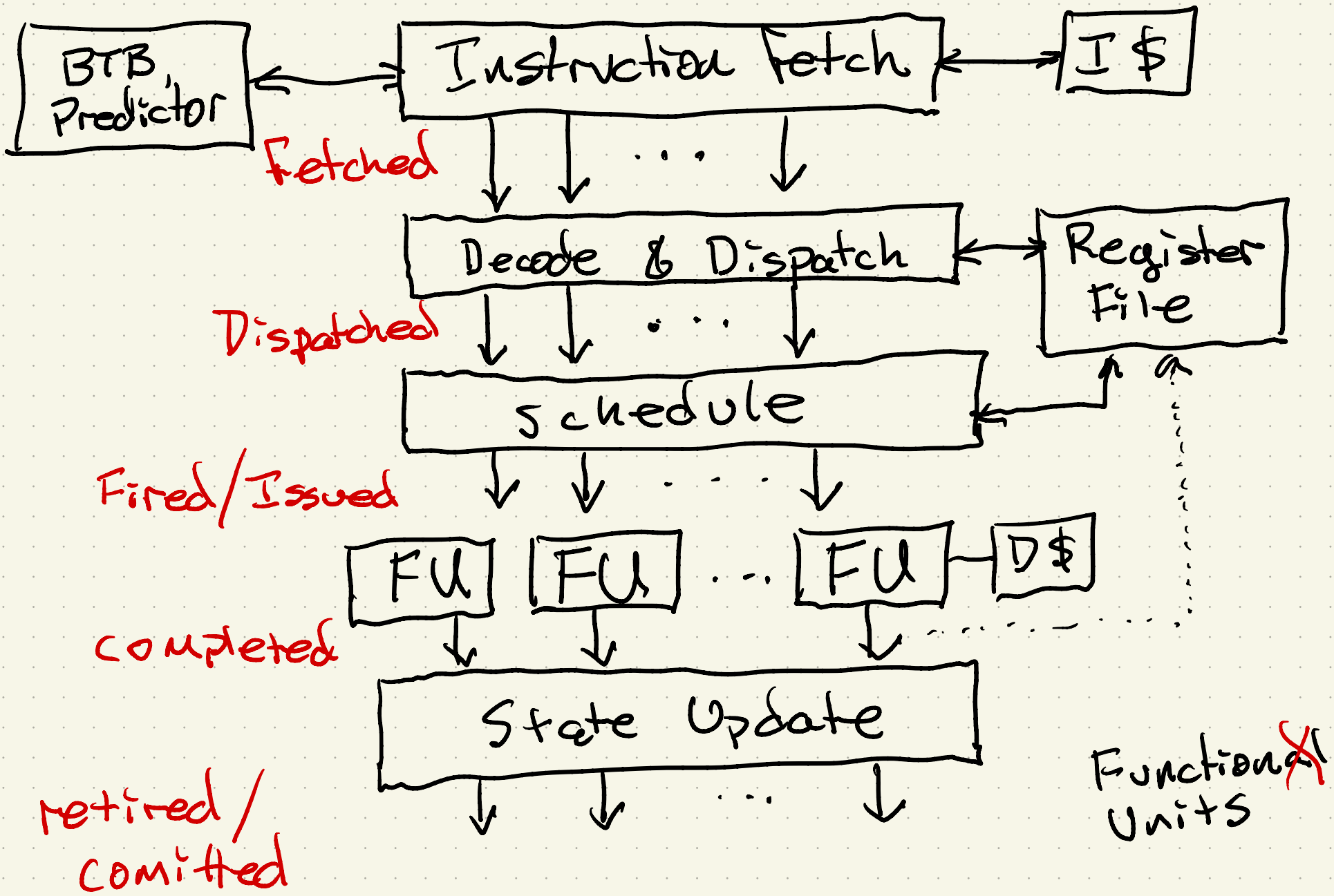


CPI ≈ 0.5

CDC 6600, IBM 360/91

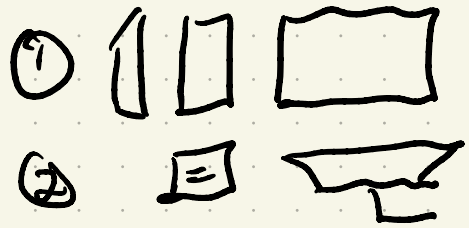
↳ Cray-1, scalar processor, supercomputer — SUPERSCALAR

Superscalar Microarchitecture



FU's

Divide? Don't care

Type	Latency	why
Integer ALU	1	ALU is critical
FP MUL	3	① Denormalize ② Add exponents, mul mantissas ③ Renormalize
FP ADD	2	① Denormalize ② Add, Renormalize
Load (LW)	2	
Store (SW)	1	Load/store buffer

Classification of Superscalar

Fire: in-order (I) / out-of-order (O)

Complete: in-order (I) / out-of-order (O)

Cray-1 superscalar

FICO

IBM 360/91

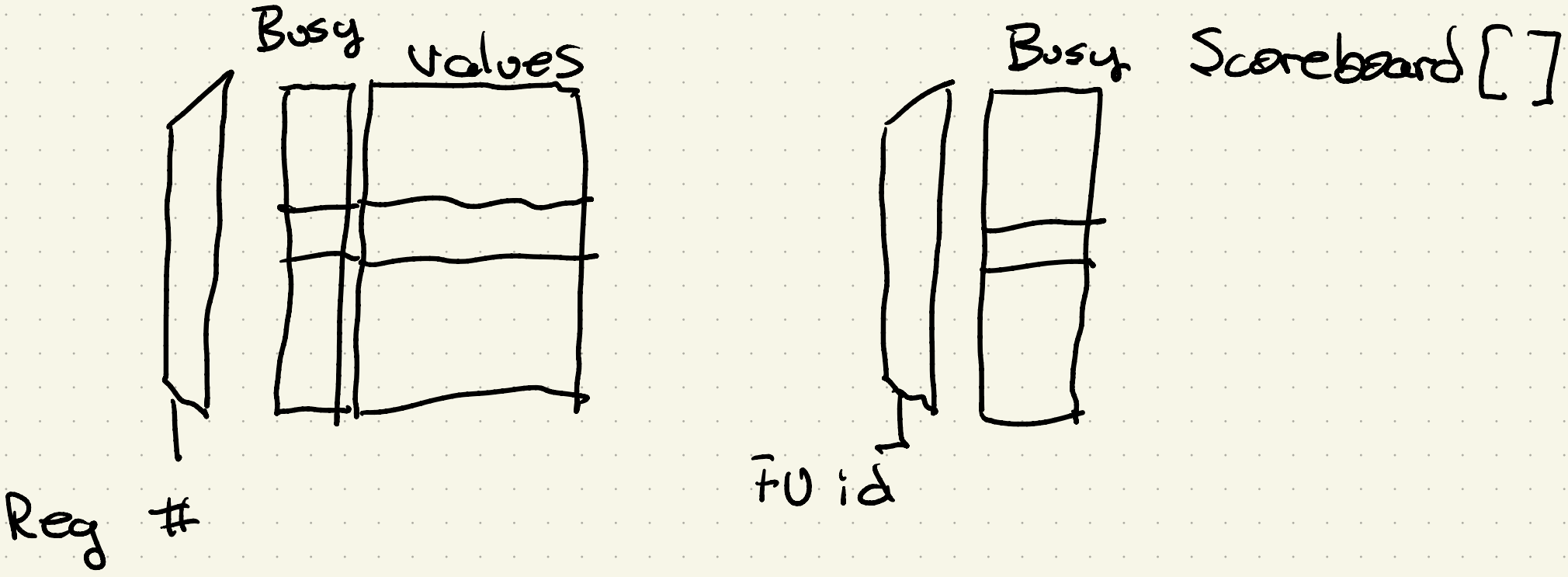
FOCO

Traditional

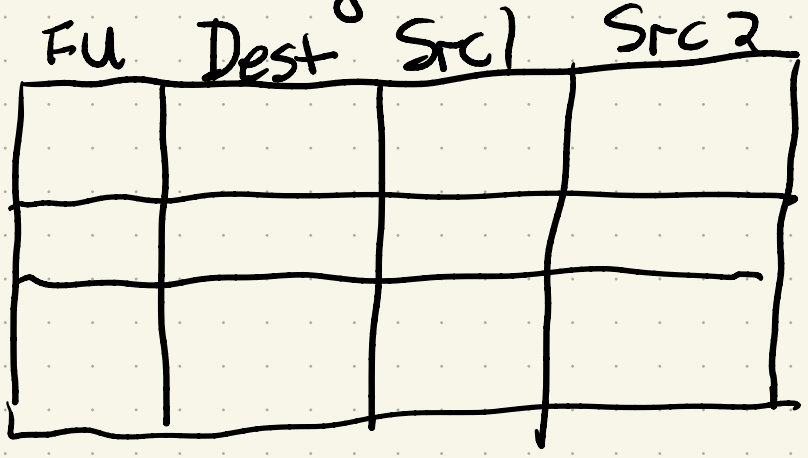
FICI

~~FOCI~~

FIFO simple interlocking



Scheduling Queue



FIFO algorithm

[Dispatch/Scheduling Unit]

```
for each Inst in SchedQ do:  // in prog. order
    if (!Regs[Inst.Src1].Busy
      AND !Regs[Inst.Src2].Busy
      AND !Regs[Inst.Dest].Busy
      AND !Scoreboard[Inst.FU].Busy)
    Then // Fire instruction
        Scoreboard[Inst.FU].Busy = True
        Regs[Inst.Dest].Busy = True
    Else
        exit loop  // halt issuing for this cycle
```

[Execution unit at completion of Inst]

Regs[Inst.Dest].Busy = False

if (Inst.FU advances and first stage is free)

Scoreboard[Inst.FU].Busy = False

FICO example

